

OWATONNA DEGNER, MN, USA

WMO: 726568

Lat: 44.123N

Lon: 93.261W

Elev: 1146

StdP: 14.10

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04961

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.6	(c) -8.6	(d) -20.4	(e) 1.9	(f) -11.7	(g) -16.6	(h) 2.3	(i) -8.3	(j) 28.8	(k) 15.6	(l) 26.7	(m) 17.9	(n) 6.3	(o) 280	(p) 0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.4	(c) 89.9	(d) 74.2	(e) 86.4	(f) 72.6	(g) 83.7	(h) 71.2	(i) 78.0	(j) 85.2	(k) 75.8	(l) 82.9	(m) 74.0	(n) 80.8	(o) 12.7	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	138.5	82.6	73.2	128.6	80.2	72.0	123.6	79.3	42.4	85.2	40.0	83.4	38.3	80.5	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 26.4	(b) 23.6	(c) 20.5	(e) -19.4	(f) 94.7	(g) 5.6	(h) 3.4	(i) -23.4	(j) 97.2	(k) -26.7	(l) 99.2	(m) -29.8	(n) 101.2	(o) -33.8	(p) 103.7		
(4) 26.4	(5) 23.6	(6) 20.5	(7) -19.8	(8) 81.4	(9) 5.5	(10) 2.6	(11) -23.8	(12) 83.2	(13) -27.0	(14) 84.7	(15) -30.1	(16) 86.2	(17) -34.1	(18) 88.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.9	15.3	19.1	32.2	46.2	58.4	68.4	71.8	69.3	62.5	49.1	35.0	21.3	(19)	(20)
	DBStd	22.31	13.41	13.04	12.67	10.29	8.97	6.74	5.68	5.62	8.74	9.60	11.34	12.28		
	HDD50	4213	1075	865	565	188	25	0	0	0	8	134	464	889		
	HDD65	7632	1540	1285	1017	569	245	41	9	22	150	500	900	1354		
	CDD50	2702	0	0	14	73	284	553	676	599	382	107	14	0		
	CDD65	647	0	0	1	4	40	144	220	156	74	8	0	0		
	CDH74	5732	0	0	5	58	447	1255	2023	1222	639	82	1	0		
	CDH80	1669	0	0	0	14	148	390	629	298	174	16	0	0		
	WSAvg	9.3	10.3	10.3	10.4	11.1	10.4	8.7	6.9	6.2	7.6	9.3	10.3	9.9		
	PrecAvg	34.3	0.9	0.7	2.1	3.0	4.1	5.1	4.4	5.0	3.5	2.4	1.7	1.1		
	PrecMax	41.1	3.0	1.8	3.8	5.8	7.6	8.6	8.4	10.3	11.2	6.1	5.2	3.1		
(15) Precipitation	PrecMin	23.0	0.2	0.1	0.4	0.7	1.6	2.7	1.3	1.7	0.9	0.5	0.2	0.3	(16)	(17)
	PrecStd	4.5	0.7	0.5	0.9	1.4	1.8	1.7	2.0	2.0	2.4	1.5	1.3	0.7		
	DB	43.4	53.7	71.7	81.5	90.6	92.4	91.9	90.1	89.9	82.0	68.4	54.4	50.0		
	MCWB	38.8	48.3	59.1	62.2	67.3	73.8	78.0	75.9	71.3	63.8	55.8	50.0	50.0		
(21) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	38.5	45.1	62.8	73.0	83.7	88.3	89.8	85.9	84.2	75.0	61.5	45.4	41.6	(22)	(23)
	2% MCWB	35.4	41.9	55.0	56.8	64.4	71.3	77.3	74.7	70.0	60.8	51.8	41.6	41.6		
	5% DB	36.2	39.1	55.2	67.9	79.1	84.0	86.1	82.5	81.4	70.2	55.4	39.4	36.6		
	5% MCWB	34.0	36.2	49.2	54.5	62.0	69.3	74.7	72.4	67.6	59.0	48.5	36.2	36.2		
	10% DB	33.8	36.4	49.6	63.3	73.3	81.3	83.5	81.1	77.3	64.2	51.6	36.6	36.6		
	10% MCWB	32.1	34.2	44.8	51.8	59.5	67.7	73.2	71.2	65.9	55.8	45.7	34.6	34.6		
	0.4% WB	39.5	48.0	62.9	65.0	72.5	78.9	82.0	79.8	76.0	69.7	58.9	50.8	50.8		
	0.4% MCDB	43.2	51.5	67.5	73.9	84.4	86.3	88.4	85.6	83.9	76.3	66.6	54.2	54.2		
	2% WB	35.9	41.6	56.2	60.5	68.3	75.3	78.9	77.1	73.2	64.4	54.3	41.5	41.5		
	2% MCDB	38.0	44.7	60.9	69.8	78.3	83.3	86.5	83.4	80.7	69.7	59.7	45.1	45.1		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5% WB	33.6	37.0	50.2	56.7	65.6	72.6	76.5	75.0	70.8	60.4	49.6	37.3	37.3	(31)	(32)
	5% MCDB	35.5	39.0	55.0	65.3	73.6	80.9	83.9	81.0	78.2	68.1	55.4	39.3	39.3		
	10% WB	31.7	34.2	44.5	53.1	62.9	70.2	74.4	73.0	68.1	56.8	45.6	34.6	34.6		
	10% MCDB	33.1	36.2	48.9	61.1	70.2	78.2	81.4	78.6	75.3	64.2	49.6	36.6	36.6		
(35) Mean Daily Temperature Range	MDBR	15.5	16.1	16.0	19.5	20.1	19.0	19.4	19.2	21.0	19.2	15.7	14.1	14.1	(36)	(37)
	5% DB	15.9	16.4	22.9	27.9	27.0	22.9	20.4	20.5	23.4	26.0	23.1	17.0	17.0		
	5% WB	14.0	13.7	15.8	15.3	12.3	10.4	11.0	11.5	11.6	14.0	15.1	14.1	14.1		
	MCDBR	14.9	14.8	21.3	24.1	21.2	19.5	18.5	17.8	19.6	20.5	20.8	15.4	15.4		
(40) Clear-Sky Solar Irradiance	taub	0.267	0.289	0.315	0.369	0.396	0.408	0.408	0.402	0.373	0.324	0.291	0.267	0.267	(41)	(42)
	taud	2.426	2.354	2.409	2.310	2.290	2.297	2.297	2.312	2.364	2.496	2.503	2.448	2.448		
	Ebn at Noon	276	287	293	284	278	274	272	270	269	272	265	265	265		
	Edh at Noon	23	30	32	39	41	41	40	38	34	26	21	20	20		
(44) All-Sky Solar Radiation	RadAvg	558	899	1182	1455	1704	1905	2011	1687	1336	863	560	407	407	(45)	(46)
	RadStd	65	95	117	155	135	152	83	118	96	115	52	35	35		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page